

CONTEXT_MAP.md

Domain Analysis & Architecture

1. Bounded Contexts

Based on the file structure and logic, the application is divided into three primary (but entangled) contexts:

- **Identity & Access Management (IAM):**
 - **Responsibilities:** User authentication, session management, and access control.
 - **Files:** `routes/login.js`, `routes/login_check.js`, `model/auth.js`.
 - **Reasoning:** These files exclusively handle login flows, session validation, and user verification against the database.
- **Product Catalog:**
 - **Responsibilities:** Displaying products, searching, and viewing details.
 - **Files:** `routes/products.js` (read methods), `model/products.js` (read methods).
 - **Reasoning:** The "Browsing" capability is distinct from the "Buying" capability, though currently mixed in the same files.
- **Order Management (Sales):**
 - **Responsibilities:** processing purchases, recording transaction details.
 - **Files:** `routes/products.js` (`/buy` endpoint), `model/products.js` (`purchase` method), `model/init_db.js` (Schema definition for `purchases`).
 - **Reasoning:** This represents the transactional side of the business.

2. Coupling Analysis (Leaky Abstractions)

- **The "Product" God-Object:** `model/products.js` is a clear violation of Single Responsibility Principle. It handles:
 1. Fetching product reference data (Catalog).
 2. Executing financial transactions/orders (Sales).
 3. Searching (Search).

- **Controller Logic Leakage:** `routes/products.js` contains significant business logic, specifically in the `/buy` endpoint. It performs validation on email formats and cart structure directly in the HTTP route handler rather than delegating to a domain service or validator.
- **Infrastructure Leakage:** SQL queries are hardcoded directly into the model functions (`model/products.js`). The database schema is defined in JavaScript (`model/init_db.js`) rather than migration files, coupling the app runtime to schema management.

3. Context Map (Mermaid)

Figure 1: Architecture Diagram — Context Map Visualization

